

Histology of Skin Lecture Notes

Reading: Gartner, “Textbook of Histology”, 5th edition, Chapter 14.

Objectives:

1. Contrast the locations and compositions of thick and thin skin.
2. Know and identify the five different layers in the epidermis of thick skin.
3. Know and identify the three different layers found in the epidermis of thin skin.
4. Understand the composition and histological organization of dermis.
5. Describe the locations and organizations of sweat glands, nerve corpuscles, and hair follicles found in the dermis.

I. Introduction

Skin is the largest organ of the body. Skin has many functions, including protection, regulation of body temperature, reception of sensations from the environment, excretion of sweat and absorption of UV radiation for Vitamin D synthesis. Skin consists of an outer epidermis and an underlying dermis. The epidermis is a stratified squamous keratinized epithelium. The dermis is a dense irregular collagenous connective tissue. The interface between the epidermis and dermis consists of undulations formed by dermal ridges (papillae) that interdigitate with epidermal ridges. The hypodermis, a loose connective tissue containing fat, lies below the dermis but is not formally part of the skin. The hypodermis is the superficial fascia of gross anatomy.

In different parts of the body, skin is different. Skin of the eyelids is thin with fine hairs. In contrast, skin of the palms of the hands and soles of the feet is very thick, lacks hair, but contains sweat glands.

II. Epidermis

The epidermis of both thin and thick skin is a stratified squamous keratinized epithelium. In thick skin, five layers of cells are present in the epidermis. From the deep to the superficial layer they are: stratum basale (also called stratum germinativum), stratum spinosum, stratum lucidum and stratum corneum (keratin). The main cell type of the epidermis is the keratinocyte. Keratinocytes are born in the stratum basale, and then progress through five morphological distinguishable stages. Concomitantly, they move upwards in the epidermis, thereby giving rise to the five different layers of the epidermis.

The stratum basale is the deepest layer of skin. It sits on a basement membrane on the dermis. The stratum basale consists of a single layer of mitotically active stem cells for keratinocytes. These cells are irregular in shape: cuboidal to low columnar. Mitosis of keratinocytes in the stratum basale is responsible for replenishment of the epithelium. Addition of cells to the stratum basale through cell division will progressively push the previous layer of keratinocytes in the stratum basale upwards to the next layer: the stratum spinosum.

The stratum spinosum contains several layers of keratinocytes that are polymorphous in shape. These keratinocytes contain numerous cellular processes containing sites of cell-cell attachment (desmosomes). In histology slides, shrinkage of the cells following fixation results in the keratinocytes pulling away from one another. The cellular processes of the keratinocytes remain attached to one another, thereby giving a spiny appearance to the cells.

The stratum granulosum is the 2-3 layers of keratinocytes just below the keratin layer. These keratinocytes are characterized by dark, keratohyalin granules in their cytoplasm. (Bundles of keratin filaments pass through these granules.) These cells of the stratum granulosum also secrete a membrane coating substance that prevents the cells lying superficial to this region from being bathed in nutrient rich extracellular fluid, facilitating their death.

The stratum lucidum is a very thin discontinuous layer of cells immediately superficial to the stratum granulosum. The flattened cells of the stratum lucidum lack organelles and functional nuclei but contain densely packed keratin filaments located parallel to the skin. They also contain eleidin, a transformation product of keratohyalin.

The stratum corneum is composed of several-many layers of flattened, dead cells containing keratin known as squames. These cells lack organelles, nuclei and are filled with keratin. The keratin gives extra protection to the skin.

In addition to the keratinocytes in the various developmental stages, the epidermis also contains scattered Melanocytes. Melanocytes are derived from the neural crest, and are found among the cells of the stratum basale. Melanocytes are round to columnar cells with long, undulating processes that penetrate up to the stratum spinosum. Melanocytes synthesize melanin and then donate it to the overlying keratinocytes in the stratum spinosum through their processes. The melanin then localizes to the supranuclear region of the keratinocytes, thereby preventing the UV radiation from reaching their nuclei. In histology slides, we observe the cell bodies of melanocytes and their dark nuclei.

In thin skin, the epidermis is identical to that of thick skin but contains only 3 of the 5 layers. Specifically, thin skin epidermis contains stratum basale, stratum spinosum, and stratum corneum.

III. Dermis

The dermis underlying the epidermis is divided into two layers- the superficial loosely woven papillary layer, and the more deep underlying layer called the reticular layer. (The bulk of the dermis is reticular). Both the papillary and reticular layers are irregular collagenous connective tissue, with the reticular layer being dense and the papillary layer being loose. The connective tissue contains mostly type I collagen, and some networks of elastic fibers. Underlying the dermis is the hypodermis-equivalent to the superficial fascia of gross anatomy.

The papillary layer of the dermis contains many capillary loops, which extend into the epidermis-dermis loops. Located in dermal papillae are pear shaped Meissner's corpuscles, mechanoreceptors specialized to respond to slight deformations of the epidermis.

The reticular layer of the dermis contains sweat glands, hair follicles and sebaceous glands. Other smooth muscle fibers called arrector pilli muscles are inserted into the hair follicle. The reticular layer also contains Pacinian corpuscles- deep pressure touch receptors. These appear like large onions due to the many, many concentric rings of supportive cells encircling the nerve endings.